

# ECVP2026 Quarto demo with knitr engine

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## Add a heading in your document

This is a sentence with some **bold text**, *italic text*, `code`, and a [link](#).



Figure 1: The Quarto logo

See Figure 1 for the Quarto logo. Equation 1 gives the formula for the population mean:

$$\mu = \frac{\sum x}{N} \quad (1)$$

Section shows how to add R or Python code chunks.

The palmerpenguins package was developed by Horst, Hill, and Gorman (2020). We will create a document using Quarto (Allaire et al. 2025) and R (R Core Team 2024) or Python (Van Rossum and Drake 2009).

This sentence ends with a footnote.<sup>1</sup>

## Add R/Python code chunks

### Add R code

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<sup>1</sup>This is an example footnote.

```
library(palmerpenguins) # for data
library(tidyverse)      # for data wrangling and visualization
library(knitr)          # for tables

library(IRdisplay)

#install.packages("palmerpenguins", repos = "http://cran.us.r-project.org")

ggplot(data = penguins,
       aes(x = flipper_length_mm, y = bill_length_mm)) +
  geom_point(aes(color = species, shape = species))
```

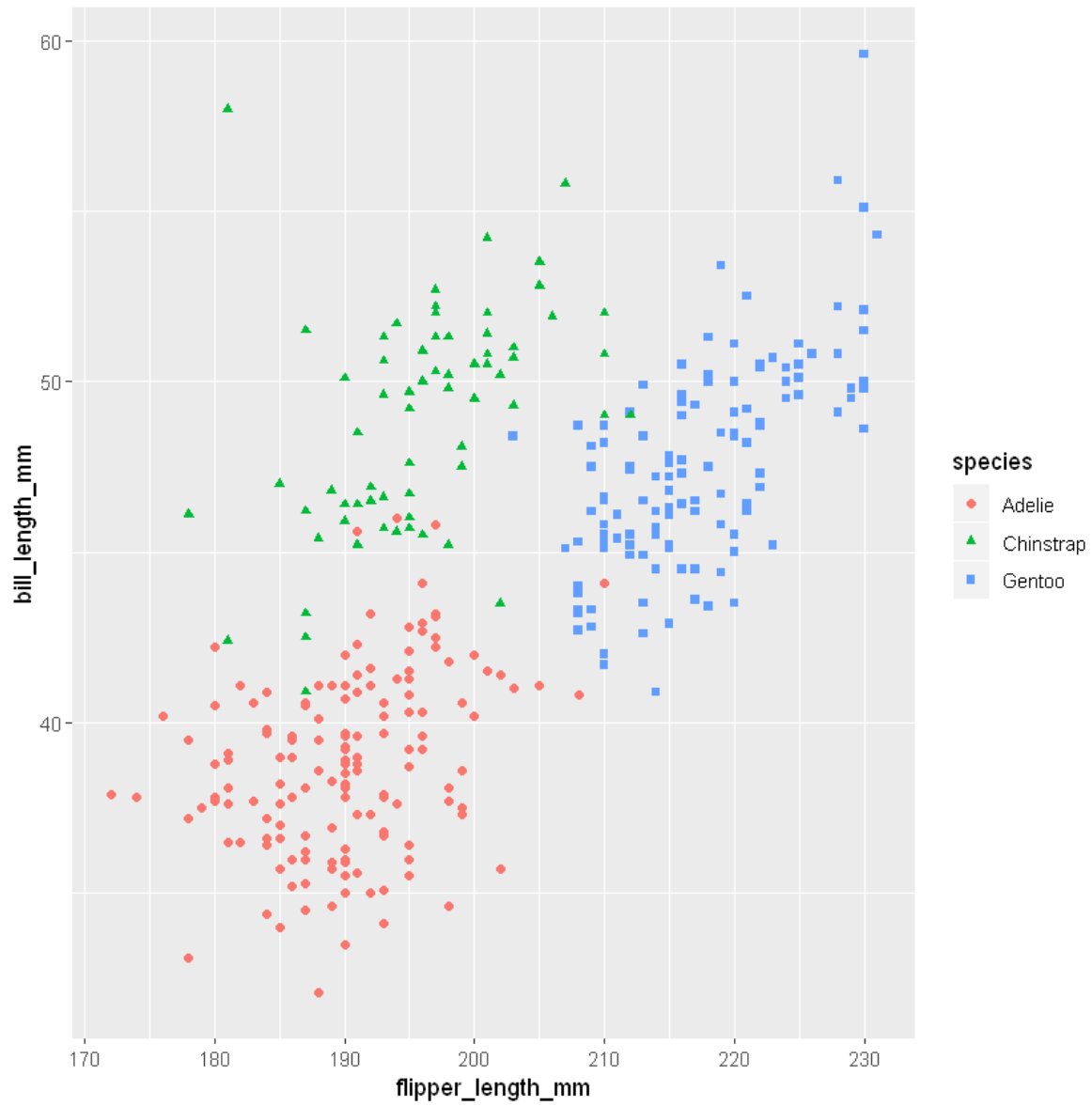


Figure 2: Scatterplot of flipper and bill lengths in R

### Add inline code

The palmerpenguins package contains data for `{r} nrow(penguins)` penguins.

## Add tables

Table 1 and Table 2, show different table options in Quarto.

## Markdown

```
| fruit | price |  
|-----|-----|  
| apple | 2.05 |  
| pear  | 1.37 |  
| orange | 3.09 |  
  
: Fruit prices {#tbl-md .striped .hover}
```

Table 1: Fruit prices

| fruit  | price |
|--------|-------|
| apple  | 2.05  |
| pear   | 1.37  |
| orange | 3.09  |

## R

```
penguins %>%  
  group_by(species) %>%  
  summarise(  
    `Mean bill length` = mean(bill_length_mm, na.rm = T) %>% round(2),  
    `Min bill length` = min(bill_length_mm, na.rm = T) %>% round(2),  
    `Max bill length` = max(bill_length_mm, na.rm = T) %>% round(2),  
    `Mean flipper length` = mean(flipper_length_mm, na.rm = T) %>% round(2),  
    `Min flipper length` = min(flipper_length_mm, na.rm = T) %>% round(2),  
    `Max flipper length` = max(flipper_length_mm, na.rm = T) %>% round(2),  
    `Correlation, r` = cor(flipper_length_mm, bill_length_mm, use = "complete") %>% round(2)  
  )
```

Table 2: Summary statistics for flipper and bill lengths

| species   | Mean bill<br>length | Min bill<br>length | Max bill<br>length | Mean<br>flipper<br>length | Min<br>flipper<br>length | Max<br>flipper<br>length | Correlation,<br>r |
|-----------|---------------------|--------------------|--------------------|---------------------------|--------------------------|--------------------------|-------------------|
| Adelie    | 38.79               | 32.1               | 46.0               | 189.95                    | 172                      | 210                      | 0.33              |
| Chinstrap | 48.83               | 40.9               | 58.0               | 195.82                    | 178                      | 212                      | 0.47              |
| Gentoo    | 47.50               | 40.9               | 59.6               | 217.19                    | 203                      | 231                      | 0.66              |

## References

- Allaire, J. J., Charles Teague, Carlos Scheidegger, Yihui Xie, Christophe Dervieux, and Gordon Woodhull. 2025. “Quarto.” <https://doi.org/10.5281/zenodo.5960048>.
- Horst, Allison M, Alison Presmanes Hill, and Kristen B Gorman. 2020. *Allison-horst/Palmerpenguins: V0.1.0*. Zenodo. <https://doi.org/10.5281/ZENODO.3960218>.
- R Core Team. 2024. *R: A Language and Environment for Statistical Computing*. Vienna, Austria: R Foundation for Statistical Computing. <https://www.R-project.org/>.
- Van Rossum, Guido, and Fred L. Drake. 2009. *Python 3 Reference Manual*. Scotts Valley, CA: CreateSpace.